

Mathematics A

Unit 2 • Chapter OL-T26 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 16 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T26 • Expertise Q16+ • 4 items • 16 marks

Chapter	Items	Marks	Starts
01. OL-T26 Solving Inequalities	4	16	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Solving Inequalities

Unit 2 • Expertise • 16 marks • 4 item(s)

June 2025 | 4MA1-2H Q19 | 3 marks

Factorable quadratic sign intervals

January 2022 | 4MA1-2H Q22 | 5 marks

Contextual quadratic feasible range

May-Nov 2020 | 2H Q17 | 5 marks

Inequality symbol from a number line

January 2023 | 2H Q22 | 3 marks

One-sided linear inequality

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Show clear algebraic working.

.....
(Total for Question 19 is 3 marks)

WORKING SPACE

Quadratic inequality

3 marks

Factorise

$$4x^2 + 4x - 15 = (2x - 3)(2x + 5).$$

Use the sign of an upward quadratic

The expression is negative between the roots $-\frac{5}{2}$ and $\frac{3}{2}$.

Final answer

$$-2.5 < x < 1.5$$

4MA1-2H Q22

5 marks

22 Here is a rectangle.

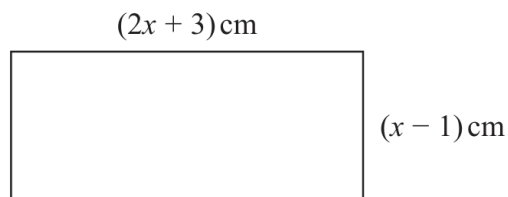


Diagram **NOT**
accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

WORKING SPACE

4MA1-2H Q22

5 marks

22 Here is a rectangle.

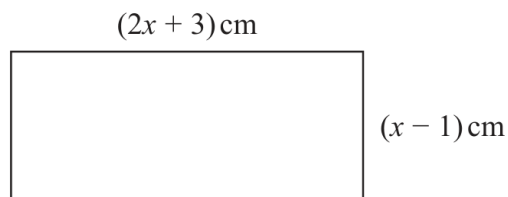


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Chapter: Solving Inequalities

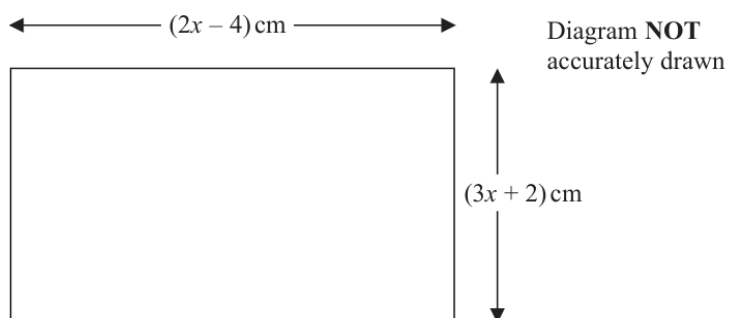
5 marks

Q22 – Chapter: Solving Inequalities

Family: Solve quadratic inequalities • Variant: Contextual quadratic feasible range

Method $(2x+3)(x-1) < 75$ gives $2x^2 + x - 78 < 0$.*Why: Expand and move 75 to the left.***Method**The critical values solve $2x^2 + x - 78 = 0 = (x-6)(2x+13)$, so $x=6$ or $x=-6.5$.*Why: Factorise the boundary equation.***Method**Because the rectangle side $x-1$ must be positive, $x > 1$.*Why: Lengths are positive.***Method**The quadratic is negative between its roots, so $x < 6$.*Why: Use the sign of the upward-opening quadratic.***Method**Therefore $1 < x < 6$.*Why: Combine the restrictions.***Final answer**range: $1 < x < 6$

17 The diagram shows a rectangle.



The area of the rectangle is $A \text{ cm}^2$

Given that $A < 3x + 27$

find the range of possible values for x .

(Total for Question 17 is 5 marks)

Answer reference | May-Nov 2020 | Question 17

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

17	$(3x+2)(2x-4) < 3x+27$ or eg $6x^2 - 8x - 8 < 3x + 27$ eg $6x^2 - 11x - 35 < 0$			M1	condone incorrect symbol
				M1	expanding and rearranging to get a correct 3 term quadratic, condone incorrect symbol
	$(2x-7)(3x+5) (= 0)$ or $\frac{11 \pm \sqrt{(-11)^2 - 4 \times 6 \times (-35)}}{2 \times 6}$			M1	first step to find the critical values dep on M1 for solving their 3 term quadratic using any correct method (allow one sign error and some simplification – allow as far as the equivalent of $\frac{11 \pm \sqrt{121+840}}{12}$) or if factorising, allow brackets which expanded give 2 out of 3 terms correct)
	$-\frac{5}{3}, \frac{7}{2}$			A1	oe the positive critical value only or both critical values (if both they must be correct)
		$2 < x < \frac{7}{2}$	5	A1	accept $2 \leq x < \frac{7}{2}$ may be seen as two separate inequalities $x > 2$ ($x \leq 2$) and $x < \frac{7}{2}$
					Total 5 marks

Worked solution

May-Nov 2020 | Question 17 | 5 marks

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

Official final mark-scheme pages are embedded immediately after this question.

- 22** Solve the inequality $6x^2 + 37x \leq 35$
Show clear algebraic working.

.....
(Total for Question 22 is 3 marks)

Answer reference | January 2023 | Question 22

Family: Solve linear inequalities • Variant: One-sided linear inequality

Question	working	Answer	Mark	Notes
22	$(6x - 5)(x + 7) = 0$ or $\frac{-37 \pm \sqrt{37^2 - 4 \times 6 \times -35}}{2 \times 6}$ $6 \left[\left(x + \frac{37}{12} \right)^2 - \left(\frac{37}{12} \right)^2 \right] \dots$ oe		3	M1 A correct method to solve the quadratic equation $6x^2 + 37x - 35 = 0$ using any correct method (if factorising, allow brackets which expanded give 2 out of 3 terms correct) (if using formula allow one sign error in substitution and some simplification – allow as far as $\frac{-37 \pm \sqrt{1369 + 840}}{12}$) or completing the square as far as shown on left
	$\frac{5}{6}$ oe and -7			A1 dep on M1 correct critical values (allow 0.83...)
	<i>Working must be seen for both accuracy marks as asked for in question</i>	$-7 \leq x \leq \frac{5}{6}$		A1 dep on M1 oe eg $-7 \leq x \leq 0.83\dots$, $\left[-7, \frac{5}{6} \right]$ Accept $x \leq \frac{5}{6}, x \geq -7$
Total 3 marks				

Worked solution

January 2023 | Question 22 | 3 marks

Family: Solve linear inequalities • Variant: One-sided linear inequality

Official final mark-scheme pages are embedded immediately after this question.